1.4 Predicates and Quantifiers

- Propositional logic cannot adequately express the meaning of all statements in mathematics and in natural language. For example, suppose that we know that

"Every computer connected to the university network is functioning properly."

No rules of propositional logic allows us to conclude the truth of the statement "MATH3 is functioning properly," where MATH3 is one of the computers connected to the university network.

Likewise, we cannot use the rules of propositional logic to conclude from the statement "CS2 is under attack by an intruder.", where CS2 is a computer on the university network to conclude that the truth of

"There is a computer on the university network that is under attack by an intruder."

For more expression of meaning of statements, we introduce Predicate Logic.